Contact Law Discovery via Symbolic Regression in Rotordynamics*

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Abstract. This study integrates symbolic regression (SR) with rotor-bearing-impact dynamics for contact force identification. A hydrodynamic Jeffcott rotor is modeled using Krämer's approximation and the Hunt & Crossley contact law with Coulomb friction. Using a physically-constrained loss function in PySR, the algorithm successfully recovers the exact Hunt & Crossley functional form and parameters (stiffness and restitution) from clean data. Under 1% Gaussian noise, the fundamental equation structure remains robust with minor parameter deviations. The results validate SR as an automated, interpretable approach for governing equation discovery in rotordynamics.

Keywords: symbolic regression · impact modeling · rotordynamics · system identification · contact mechanics · hydrodynamic bearings

1 Introduction

Rotor-bearing-impact analysis involves complex nonlinear interactions where direct measurement of contact forces and indentation depths is often hindered by sensor limitations and noise. To simulate these dynamics, this study employs a Jeffcott rotor supported by hydrodynamic bearings, modeled via Krämer's short-bearing approximation [1] based on DuBois and Ocvirk's solution [2]. This setup captures essential behaviors such as unbalance response and oil whirl. For impact dynamics, the Hunt & Crossley model [3] is selected over simpler formulations due to its ability to represent non-linear damping and energy dissipation consistent with experimental data [4].

Symbolic regression (SR) offers a powerful automated alternative for discovering governing equations from data. Evolving from early heuristic systems [5,6] and genetic algorithms [7], modern SR tools allow for the extraction of physical invariants from noisy datasets [8,9]. Recent applications have demonstrated SR's versatility, ranging from the rediscovery of fundamental laws [10] to industrial process modeling [11].

The novelty of this work consists in the application of symbolic regression specifically to contact problems within the field of rotordynamics. While SR

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has been used to identify general dynamical systems, its use to distinguish and formulate the specific non-linearities of impact events in rotating machinery remains unexplored. Bridging these domains, this paper utilizes PySR [12] with a custom physically-constrained loss function to analyze data generated by a Lagrange-based rotor model. This framework aims to demonstrate SR's capability to accurately recover both the functional form and physical parameters of contact models, providing an interpretable and automated system identification method that overcomes the limitations of traditional modeling in rotordynamics.

2 Methodology

This section details the rotor-bearing-impact model. First, a Jeffcott-like rotor with hydrodynamic bearings is derived using Lagrange's formulation. The bearing forces are modeled analytically using the short-bearing approximation of Krämer, yielding frequency-dependent stiffness and damping coefficients. Subsequently, an impact model based on the Hunt & Crossley contact law and Coulomb friction is introduced to simulate the disk's interaction with a rigid barrier. Finally, the symbolic regression (SR) implementation using PySR is described, outlining the algorithm's search for interpretable governing equations from simulated data, guided by a custom loss function that enforces physical consistency. The source code and implementation details for the models and algorithms used are publicly available at `https://github.com/viniciuserra/Contact-Law-Discovery-via-Symbolic-Regression-in-Rotordynamics`.

2.1 Rotordynamic Model

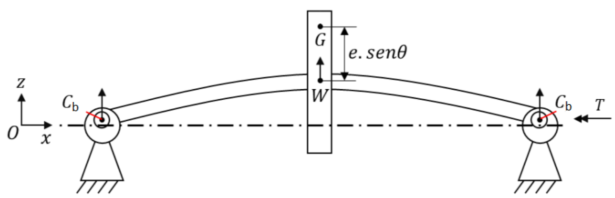
The equations of motion for the rotor system supported by hydrodynamic bearings (Fig. 1a) are derived using Lagrange's formulation. For each generalized coordinate q_l, considering kinetic energy T, potential energy V, and non-conservative generalized forces Q_l^{NC}, Lagrange's equation is:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_l}\right) - \frac{\partial T}{\partial q_l} + \frac{\partial V}{\partial q_l} = Q_l^{\mathrm{NC}} \tag{1}$$

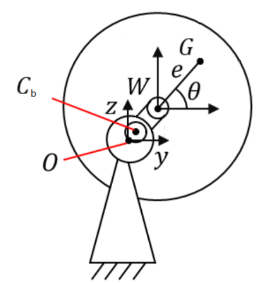
The specific equations of motion for this system are obtained by evaluating Eq. (1) for each generalized coordinate, using the energy expressions and external forces defined below.

System description The analyzed rotor, similar to a Jeffcott (Laval) rotor, consists of a disk, a shaft, and two hydrodynamic bearings. The shaft is considered elastic, massless, with constant circular cross-section, isotropic, and endowed with stiffness. The disk has mass m and is mounted at the mid-span between the two supports such that its plane remains perpendicular to the shaft. The disk center of mass (G) is non-coincident with its geometric center (W), presenting an eccentricity e (Fig. 1b).

Each hydrodynamic bearing consists of a massless journal of radius R and a lubricant of viscosity η. The journal corresponds to the portion of the shaft within the bearing, rotating with angular velocity $\dot{\theta} = \Omega$. The radial clearance between the journal and the bearing is δ_r. The bearing center is taken as the coordinate origin. The journal center (C_b) has coordinates y_b and z_b, which define the journal eccentricity $e_b = \sqrt{y_b^2 + z_b^2}$, representing the radial displacement of the journal center relative to the bearing center. Meanwhile, the disk geometric center (W) has coordinates u_y and u_z.



(a) Front view.



(b) Side view.

Fig. 1: Rotor views supported on hydrodynamic bearings.

Bearing model The bearing forces follow the short-bearing approach by Krämer [1], based on DuBois and Ocvirk [2], yielding analytical stiffness and damping coefficients. Defining the static load F_0 and the reference force F_η:

$$F_\eta = \frac{\eta \Omega R L^3}{\delta_r^2} \tag{2}$$

where η represents the dynamic viscosity, Ω denotes the shaft rotational speed, R the journal radius, L the bearing width, and δ_r the radial clearance. The classical (S_0) and modified (S^*) Sommerfeld numbers are defined as:

$$S_0 = \frac{F_0}{2R^3 L \eta \Omega} \delta_r^2, \quad S^* = \frac{F_0}{F_\eta} = S_0 \left(\frac{2R}{L} \right)^2 \tag{3}$$

The dimensionless eccentricity $\varepsilon = e_b/\delta_r$ is obtained iteratively from the relation:

$$S^* = \frac{\pi}{2} \frac{\varepsilon}{(1-\varepsilon^2)^2} \sqrt{1 - \varepsilon^2 + \left(\frac{4}{\pi} \varepsilon \right)^2} \tag{4}$$

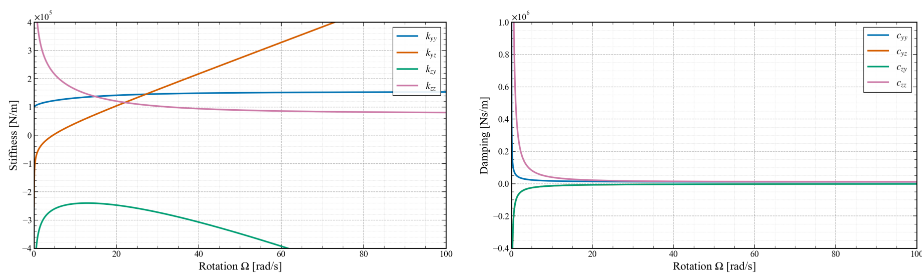
The bearing stiffness k_{ij} and damping c_{ij} coefficients are then:

$$k_{ij} = \gamma_{ij}\frac{F_0}{\delta_r}, \quad c_{ij} = \beta_{ij}\frac{F_0}{\delta_r \Omega} \tag{5}$$

where γ_{ij} and β_{ij} are dimensionless functions of ε:

$$\begin{aligned}
\gamma_{yy} &= \left[2\pi^2 + \left(16-\pi^2\right)\varepsilon^2\right] A(\varepsilon) & \beta_{yy} &= \frac{\pi\sqrt{1-\varepsilon^2}}{2\varepsilon}\left[\pi^2 + \left(\pi^2-16\right)\varepsilon^2\right] A(\varepsilon)\\
\gamma_{yz} &= \frac{\pi}{4}\frac{\pi^2 - 2\pi^2\varepsilon^2 - \left(16-\pi^2\right)\varepsilon^4}{\varepsilon\sqrt{1-\varepsilon^2}} A(\varepsilon) & \beta_{yz} &= \beta_{zy} = -\left[2\pi^2 + \left(4\pi^2-32\right)\varepsilon^2\right] A(\varepsilon)\\
\gamma_{zy} &= -\frac{\pi}{4}\frac{\pi^2 + \left(32+\pi^2\right)\varepsilon^2 + \left(32-2\pi^2\right)\varepsilon^4}{\varepsilon\sqrt{1-\varepsilon^2}} A(\varepsilon) & \beta_{zz} &= \frac{\pi}{2}\frac{\pi^2 + \left(48-2\pi^2\right)\varepsilon^2 + \pi^2\varepsilon^4}{\varepsilon\sqrt{1-\varepsilon^2}} A(\varepsilon)\\
\gamma_{zz} &= \frac{\pi^2 + \left(32+\pi^2\right)\varepsilon^2 + \left(32-2\pi^2\right)\varepsilon^4}{1-\varepsilon^2} A(\varepsilon) & A(\varepsilon) &= \frac{4}{\left[\pi^2 + \left(16-\pi^2\right)\varepsilon^2\right]^{3/2}}
\end{aligned} \tag{6}$$

The resulting stiffness and damping coefficients versus speed are illustrated in Fig. 2.



(a) Stiffness coefficients. (b) Damping coefficients.

Fig. 2: Stiffness and damping coefficients as a function of rotation speed.

Energy equations and generalized forces The system relies on five generalized coordinates: disk geometric center (u_y, u_z), journal center (y_b, z_b), and angle θ. The disk center of mass coordinates are:

$$y_G = u_y + e\cos\theta, \quad z_G = u_z + e\sin\theta \tag{7}$$

The kinetic energy T (translational and rotational, with inertia J_G) and the potential energy V (elastic and gravitational) are defined as:

$$T = \frac{1}{2}m\left[(\dot{u}_y - e\dot{\theta}\sin\theta)^2 + (\dot{u}_z + e\dot{\theta}\cos\theta)^2\right] + \frac{J_G\dot{\theta}^2}{2} \tag{8}$$

$$V = \frac{1}{2}k(u_y - y_b)^2 + \frac{1}{2}k(u_z - z_b)^2 + mg(u_z + e\sin\theta) \tag{9}$$

Non-conservative generalized forces Q_l^{NC} arise from shaft viscous damping c, applied motor torque τ, hydrodynamic bearing forces, and contact forces upon barrier impact.

The virtual work of non-conservative forces is:

$$\delta W = -c\dot{u}_y\delta u_y - c\dot{u}_z\delta u_z + T\delta\theta + Q_{y_b}\delta y_b + Q_{z_b}\delta z_b \tag{10}$$

with bearing-related generalized forces:

$$\begin{aligned} Q_{y_b} &= 2(k_{yy}y_b + k_{yz}z_b + c_{yy}\dot{y}_b + c_{yz}\dot{z}_b) \\ Q_{z_b} &= 2(k_{zy}y_b + k_{zz}z_b + c_{zy}\dot{y}_b + c_{zz}\dot{z}_b) \end{aligned} \tag{11}$$

Equations of motion Applying Lagrange's equation (1) with Eqs. (8)–(11):

$$m\ddot{u}_y + c\dot{u}_y + k(u_y - y_b) = me(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta) \tag{12}$$

$$m\ddot{u}_z + c\dot{u}_z + k(u_z - z_b) = me(-\ddot{\theta}\cos\theta + \dot{\theta}^2\sin\theta) - mg \tag{13}$$

$$\begin{aligned} J_G\ddot{\theta} = \tau &- k(u_y - y_b)(e\sin\theta) + k(u_z - z_b)(e\cos\theta) \\ &- c\dot{u}_y(e\sin\theta) + c\dot{u}_z(e\cos\theta) \end{aligned} \tag{14}$$

$$k(y_b - u_y) + 2(k_{yy}y_b + k_{yz}z_b + c_{yy}\dot{y}_b + c_{yz}\dot{z}_b) = 0 \tag{15}$$

$$k(z_b - u_z) + 2(k_{zy}y_b + k_{zz}z_b + c_{zy}\dot{y}_b + c_{zz}\dot{z}_b) = 0 \tag{16}$$

The parameters used in the study are presented in Table 1.

Table 1: Parameters of the rotor with hydrodynamic bearings.

Parameter	**Value**	**Parameter**	**Value**
Shaft diameter (d_e)	10 mm	Young's Modulus (E)	200 GPa
Disk diameter (d_d)	100 mm	Density (ρ)	7850 kg/m^3
Shaft length (l_e)	900 mm	Poisson's ratio (ν)	0.3
Disk length (l_d)	20 mm	Damping ratio (c/k)	10^{-4} s
Bearing diameter ($2R$)	30 mm	Unbalance (me)	$1.5 \cdot 10^{-4}$ kg·m
Bearing width (L)	18 mm	Viscosity (η)	0.08 Pa·s
Radial clearance (δ_r)	100 μm		

Contact with a barrier Interaction with a barrier above the disk introduces a downward normal force F_N and dynamic friction μF_N, modifying Eqs. (12) and (13) to:

$$m\ddot{u}_y + c\dot{u}_y + k(u_y - y_b) = me(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta) + \mu F_N \tag{17}$$

$$m\ddot{u}_z + c\dot{u}_z + k(u_z - z_b) = me(-\ddot{\theta}\cos\theta + \dot{\theta}^2\sin\theta) - mg - F_N \tag{18}$$

2.2 Impact Model

The contact force follows the Hunt & Crossley model [3], incorporating energy dissipation via a non-linear damping term. The normal force F_N and the hysteretic coefficient χ are defined as:

$$F_N = K\delta^{3/2} + \chi\delta^{3/2}\dot{\delta}, \qquad \chi = \frac{3}{2}(1 - e_r)\frac{K}{v_{\text{approach}}} \tag{19}$$

where K is the Hertzian stiffness, δ indentation depth, $\dot{\delta}$ indentation velocity, e_r the coefficient of restitution, and v_{approach} the impact velocity. This results in the hysteresis loop shown in Fig. 3.

For a planar barrier located at a clearance $\sigma_b = 2$ mm above the disk (diameter d_d), the contact kinematics are defined as $\delta = u_z - \sigma_b$ and $\dot{\delta} = \dot{u_z}$. Simulation parameters are based on experimental data [4] with fixed values $e_r = 0.76$ and $K = 2.4614 \times 10^8\,\mathrm{N/m}^{1.5}$.

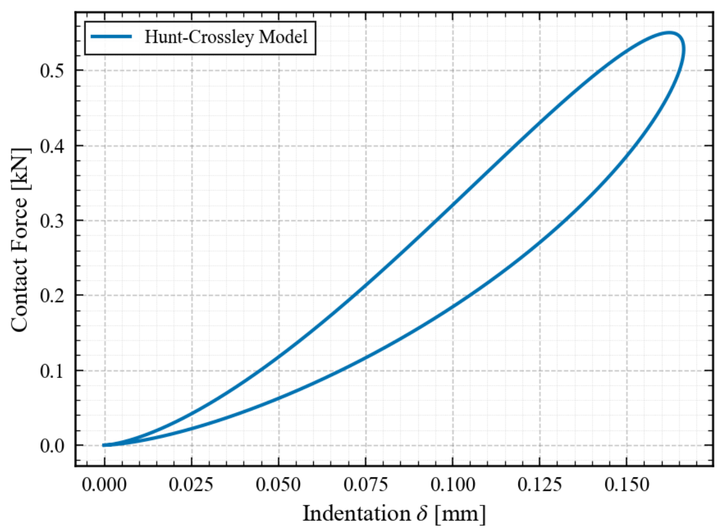


Fig. 3: Characteristic force-indentation hysteresis loops generated by the Hunt & Crossley model.

Tangential interaction is modeled using Coulomb dry friction [13], describing static and dynamic states based on the relative tangential velocity v_{rel} and coefficients μ_s, μ_d:

$$F_\mu = \begin{cases} \leq \mu_s F_N & v_{rel} = 0 \\ \mu_d F_N \,\mathrm{sgn}(v_{rel}) & v_{rel} \neq 0 \end{cases} \tag{20}$$

2.3 Symbolic Regression Implementation

We leverage Symbolic Regression (SR) to discover governing contact equations. Unlike traditional regression, SR searches the space of analytic expressions. As summarized by Cranmer (2023):

> "SR describes a supervised learning task where the model space is spanned by analytic expressions. ... In the history of science, scientists have often performed SR 'manually', using a combination of their intuition and trial-and-error to find simple and accurate empirical expressions. ... SR algorithms automate this discovery of empirical equations."[12]

We utilized PySR [12], a library employing genetic algorithms to optimize the trade-off between model accuracy and simplicity. To ensure physical validity, the optimization minimizes a composite loss function containing the Mean Squared Error (MSE) and specific physical penalties:

- **Zero-force at zero-indentation:** Penalizes non-physical forces when indentation is effectively zero ($\delta = 0$).
- **Energy conservation ($e = 1$):** Penalizes hysteresis in perfectly elastic collisions, ensuring non-dissipative behavior (null area under the curve in Fig. 3).

The total loss function $\mathcal{L}$ is defined as:

$$\mathcal{L} = \underbrace{\frac{1}{N}\sum_{i=1}^{N}(F_i - \hat{F}_i)^2}_{\text{MSE}} + \lambda_1 \sum |\hat{F}(\delta = 0)| + \lambda_2 \sum |\mathcal{H}_{e=1}| \tag{21}$$

where F_i and $\hat{F}_i$ are the measured and predicted forces, $\mathcal{H}_{e=1}$ represents a term to refer to hysteresis dissipation calculated for cases where $e = 1$, and λ_1, λ_2 are the regularization weights.

3 Results

The equations of motion were solved using Python's `solve_ivp` with the BDF method. Simulation parameters follow Section 2.1, incorporating the Hunt & Crossley contact model [3] with Hertzian stiffness $K = 2.4614 \times 10^8\,\mathrm{N/m}^{1.5}$ and restitution coefficient $e_r = 0.76$.

First, the system's free acceleration under constant torque ($\tau = 0.015\,\mathrm{N}\cdot\mathrm{m}$) was analyzed without contact (Fig. 4). Subsequently, a flat barrier was introduced at $z = d_d/2 + \sigma_b$, where $\sigma_b = 2$ mm. Figures 5 and 6 illustrate the impact dynamics, where v_{approach} is updated instantaneously prior to each impact.

3.1 Symbolic Regression for Contact Force Identification

Symbolic regression (SR) was applied to both clean and noisy data (1% Gaussian noise) to recover the contact equations. The hysteresis penalty in Eq. (21) was omitted as e_r remained constant ($e_r = 0.76$).

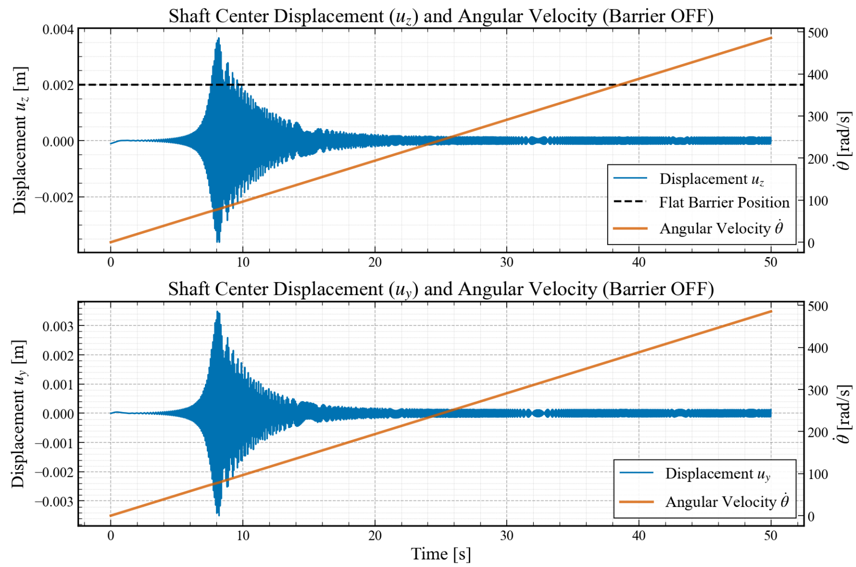


Fig. 4: Shaft center displacements and angular velocity ($\tau = 0.015\,\mathrm{N \cdot m}$) without contact. Dashed line: potential barrier position.

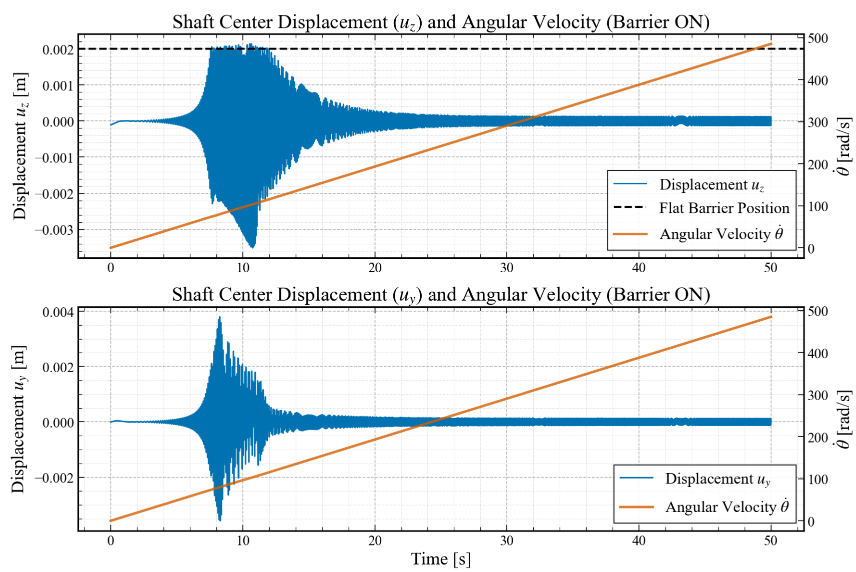


Fig. 5: Shaft center displacements and angular velocity with contact interaction at $z = 2$ mm.

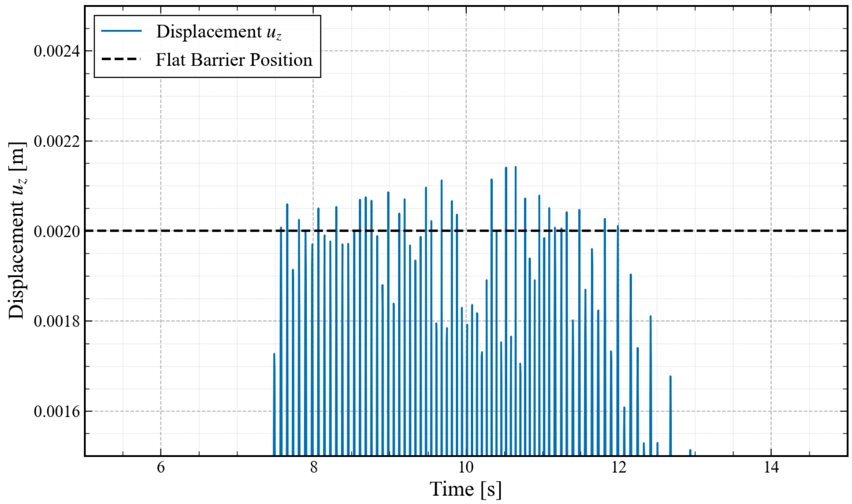


Fig. 6: Detailed view (zoom) of the shaft center displacements during contact.

Analysis with Clean Data Using inputs δ, $\dot{\delta}$, and v_{approach}, the algorithm converged to the exact Hunt & Crossley form:

$$F_N = (142.08 + 51.15\frac{\dot{\delta}}{v_{\text{approach}}}) \times (14,424.65 \times \delta)^{1.5} \tag{22}$$

Mapping this to the theoretical model ($F_N = K\delta^{1.5} + \frac{3K(1-e)}{2v}\delta^{1.5}\dot{\delta}$), the identified parameters are:

- **Stiffness:** Identified $K = 2.4600 \times 10^8$ (Target: $2.4614 \times 10^8\,\mathrm{N/m^{1.5}}$).
- **Restitution:** Identified $e = 0.7599$ (Target: $e_r = 0.76$).

This demonstrates excellent agreement, validating the algorithm's ability to retrieve physical parameters from clean data.

Analysis with Noisy Data With 1% noise added to all variables, the algorithm converged to:

$$F_N = \left(2.7659 + \frac{\dot{\delta}}{v_{\text{approach}}}\right) \times (2.0182 \times 10^5 \times \delta)^{1.4916} \tag{23}$$

Despite a slight deviation in the exponent (1.4916 vs 1.5), the structural form remained robust. The identified parameters were $K = 2.508 \times 10^8\,\mathrm{N/m^{1.5}}$ and $e = 0.759$, confirming the method's robustness against measurement noise.

4 Conclusions

This study integrated classical rotordynamics with symbolic regression to analyze rotor-bearing-impact systems. Simulations confirmed the rotor can cross its critical speed even under impact conditions. Furthermore, symbolic regression proved powerful for system identification, recovering the functional form and physical parameters of the Hunt & Crossley model from data with up to 1% noise.

By bridging mechanics and machine learning, this methodology offers a pathway toward automated and interpretable system identification. It demonstrates that data-driven approaches can effectively complement traditional analysis, reducing the need for manual model development while maintaining physical interpretability. Finally, the authors are currently working on extending this framework to applications involving real experimental data.

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